

JEE-Main-22-01-2026 (Memory Based)

[MORNING SHIFT]

Physics

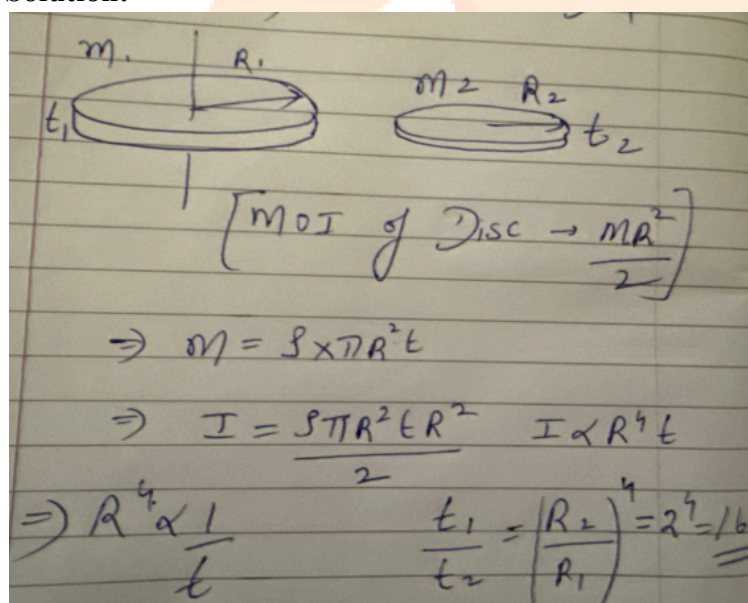
Question: Two disc having same moment of inertia about their axis. Thickness is t_1 and t_2 and they have same density. If $R_1/R_2 = 1/2$, then find t_1/t_2 .

Options:

- (a) 8
- (b) $1/8$
- (c) 16
- (d) 24

Answer: (c)

Solution:



Question: In series R-L circuit, voltage of battery is 10 V. Resistance and inductance are 10Ω and 10 mH respectively. Find energy stored in the inductor when current reaches

$\frac{1}{e}$

times of maximum value

Options:

- (a) 1.67 mJ
- (b) 0.33 mJ
- (c) 0.67 mJ
- (d) 1.8 mJ

Answer: (c)

Solution:

Ans:- Max. Current in the circuit
will be $i_0 = \frac{V}{R} = (1A)$

$$\begin{aligned}\text{Energy stored} &\Rightarrow \frac{1}{2} Li^2 \\ &\Rightarrow \frac{1}{2} \times 10^{-3} \times (1)^2 \\ &\Rightarrow \left(5 \times 10^{-3} \times \frac{1}{7.38} \right) J \\ &\Rightarrow 0.677 \times 10^{-3} J \\ &\Rightarrow 0.678 \text{ mJ}.\end{aligned}$$

Question: A particle is projected at an angle of 60° with ground when projectile makes an angle of 45° with horizontal its speed because 20 m/s. What is the initial velocity?

Options:

- (a) $3\sqrt{2}$
- (b) $20\sqrt{2}$
- (c) $5\sqrt{3}$
- (d) $5\sqrt{5}$

Answer: (b)

Solution:

Question: Find the energy to excite electron from first Bohr orbit of hydrogen atom to 2nd Bohr's orbit. (in eV)

Options:

- (a) 13.6 eV
- (b) 3.4 eV
- (c) 12.1 eV
- (d) 10.2 eV

Answer: (d)

Solution:

$$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

$$E_1 (\text{first orbit}) = E_1 = -13.6 \text{ eV}$$

$$E_2 (n=2) = E_2 = -13.6 \times \frac{1^2}{2^2}$$

$$= -3.4 \text{ eV}$$

$$\Delta E = E_2 - E_1 = 10.2 \text{ eV}$$

Question: Match the following:-

LIST-1	LIST-2
a) Spring constant	i) $\text{ML}^2\text{T}^{-2}\text{K}^{-1}$
b) Thermal conductivity	ii) ML^0T^{-2}
c) Boltzmann constant	iii) $\text{ML}^2\text{T}^{-2}\text{A}^{-2}$
d) Inductance	iv) $\text{MLT}^{-3}\text{K}^{-1}$

Options:

- (a) a-ii, b-iv, c-i, d-iii
- (b) a-i, b-ii, c-iii, d-iv
- (c) a-iv, b-iii, c-ii, d-i
- (d) a-ii, b-i, c-iv, d-iii

Answer: (a)

Solution:

Ans:- (a) Spring Const (K) = $\frac{F}{x} \Rightarrow \frac{MLT^{-2}}{L} \Rightarrow ML^{-1}T^{-2}$

(b) Thermal Conductivity $K = \frac{\left(\frac{dq}{dt}\right) \times l}{A \Delta \theta} \Rightarrow \frac{\frac{MLT^{-2}}{T} \times L}{L^2 K} \Rightarrow \frac{ML^3T^{-3} \times L^{-2}K^{-1}}{MLT^{-3}K^{-1}}$

(c) Boltzman Const :- $U = \frac{1}{2} f KT \Rightarrow \frac{U}{T} = K \Rightarrow \frac{MLT^{-2}}{K} \Rightarrow MLT^{-2}K^{-1}$

(d) Inductance $\Rightarrow U = \frac{1}{2} Li^2 \Rightarrow \frac{U}{i^2} \Rightarrow \frac{MLT^{-2}}{A^2} \Rightarrow MLT^{-2}A^{-2}$

Question: In a potentiometer null point for two resistance R_1 and R_2 is at 40 cm as shown. If 16Ω is connected in parallel to R_2 then null point is at 50 cm then R_1 and R_2 are respectively.

Options:

- (a) 16Ω , 48Ω
- (b) 32Ω , $32/3 \Omega$
- (c) $16/3 \Omega$, 8Ω
- (d) $32/3 \Omega$, 32Ω

Answer: (c)

Solution:

Case-I

$$\frac{R_1}{\cancel{240}} = \frac{R_2}{\cancel{80}}$$

$$3R_1 = 2R_2 \quad R_1 = \frac{2R_2}{3}$$

Case-II

R_2 is connected with 16 parallel

$$\frac{R_2 \times 16}{R_2 + 16} = \frac{R_1}{50}$$

$$\frac{16R_2}{R_2 + 16} = \frac{2R_2}{3}$$

$$48 = 2R_2 + 32$$

$R_1 = 16 \quad R_2 = 8 \Omega$

Question: An equilateral prism is placed in air. A ray of light is incident on one face of the prism. The minimum angle between the incident ray and the emergent ray is 90° . If the refractive index of the prism is $\sqrt{2}$, find the angle of incidence.

Options:

- (a) 50°
- (b) 90°
- (c) 60°
- (d) 75°

Answer: (d)

Solution:

$A = 60^\circ$
 angle betⁿ i & $e = 90^\circ$
 $n_i (\mu) = \sqrt{2}$

$\rightarrow \delta = 180 - 90 = 90^\circ$

$\delta = i + e - A$

for air $\rightarrow$ prism $\rightarrow$ air, (symmetry)
 $i = e$

Hence, $\delta = 2i - A$

So, $90 = 2i - 60^\circ$
 $2i = 50$
 $i = 75^\circ$

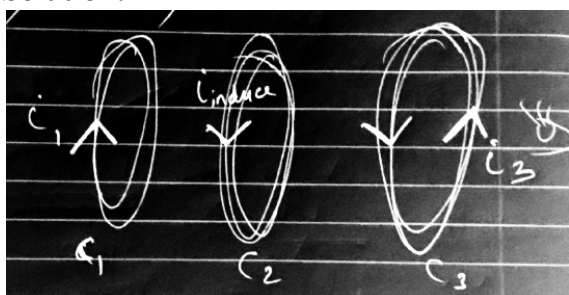
Question: Three circular coils C_1 , C_2 and C_3 are placed coaxially. When viewed from the same side, coil C_1 carries a clockwise current and coil C_3 carries an anticlockwise current. Which of the following motions of coils C_1 and C_3 will induce an anticlockwise current in the middle coil C_2 ?

Options:

- (a) Move C_1 away from C_2 and move C_3 towards C_2
- (b) Move C_1 towards C_2 and move C_3 away from C_2
- (c) Move both C_1 and C_3 towards C_2
- (d) Move both C_1 and C_3 away from C_2

Answer: (b)

Solution:



Q8 solution

Pick diagram from here for the question as well

Solution:

Moving C_1 towards C_2 increases inward flux - C_2 induces anticlockwise current to oppose it.

Coil C_3 carries anticlockwise current $\rightarrow$ magnetic field out of the page.

Moving C_3 away from C_2 decreases outward flux $\rightarrow C_2$ induces anticlockwise current to maintain it.

Hence option B is correct

01:27 PM

Question: If escape velocity of a planet is 10 km/s and for another planet which has density and radius 10% of the previous planet then its escape velocity is?

Options:

- (a) 10
- (b) $V_e/10$
- (c) 11
- (d) $V_e/11$

Answer: (b)

Solution:

Ans:- Escape velocity $V_e = \sqrt{2gR}$

$$M' = \rho' \times \frac{4}{3} \pi R'^3 ; \quad \rho' = 0.1\rho$$

$$M' = 0.1\rho \times \frac{4}{3} \pi \times (0.1R)^3 \quad R' = 0.1R$$

$$\Rightarrow 0.1\rho \times \frac{4}{3} \pi \times R^3 \times 10^{-3}$$

$$\Rightarrow 10^{-9} \times \rho \times \frac{4}{3} \pi R^3 \Rightarrow 10^{-9} \times M$$

$$g = \frac{GM}{R^2} \Rightarrow g' = \frac{GM'}{R'^2}$$

$$g' \Rightarrow \frac{G \times 10^{-9} \times M}{10^{-2} \times R^2}$$

$$g' \Rightarrow 10^{-2} \left(\frac{GM}{R^2} \right)$$

$$\frac{V_e}{V_{e'}} = \frac{\sqrt{2gR}}{\sqrt{2 \times 10^{-2} \times gR}}$$

$$\frac{V_e}{V_{e'}} = \frac{1}{\sqrt{10^{-2}}} \Rightarrow 10$$

$$\boxed{V_{e'} = \frac{V_e}{10}}$$

Question: A convex lens of focal length $f_1 = +10$ cm, $f_2 = -20/3$ cm.

Case-1:

When the two lenses are in contact, an object is placed at a distance of 10 cm from the lens combination.

The magnification in this case is M_1 .

Case-2:

Now the concave lens is shifted 1 cm away from the convex lens along the principal axis.

The object position remains unchanged (10 cm from the convex lens).

The magnification in this case is M_2 .

Find the ratio M_1/M_2 .

Options:

- (a) $3/4$
- (b) $4/3$
- (c) $2/3$
- (d) $3/2$

Answer: (b)

Solution:

$$\text{Soln } \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{10} - \frac{3}{20} = -\frac{1}{20} \quad F = -20 \text{ cm}$$

$$\text{case 1 N.W., } u = -10 \text{ cm} \quad \frac{1}{v} - \frac{1}{u} = \frac{1}{F} \Rightarrow \frac{1}{v} + \frac{1}{10} = -\frac{1}{20}$$

$$\frac{1}{v} = -\frac{3}{20} \text{ or } v = -\frac{20}{3} \text{ cm}$$

$$M_1 = \frac{v}{u} = \frac{-20/3}{-10} = \frac{2}{3}$$

$$\text{case 2 } \frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_1} \Rightarrow \frac{1}{v_1} + \frac{1}{10} = \frac{1}{10} \Rightarrow v_1 = \infty$$

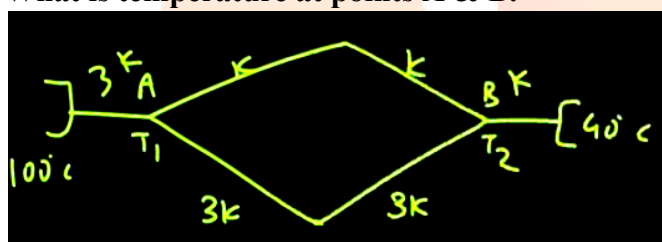
$$\text{So } v_2 = f_2 = -20/3 \text{ cm}$$

$$\text{Total mag - convex lens} = \infty / f_{\text{in}} \Rightarrow 1$$

$$\text{but because of separation } M_2 = 1/2$$

$$\text{Hence } M_1/M_2 = \frac{2/3}{1/2} = 4/3$$

Question: As shown in fig. Six rods of same geometry & maintained with 100°C & 40°C . What is temperature at points A & B.



Options:

- (a) 48°C & 52°C
- (b) 89.1°C & 72.72°C
- (c) 52.5°C & 50°C
- (d) 85°C & 72°C

Answer: (b)

Solution:

Soln

A) 48°C & 52°C
 ✓ B) 89.1°C & 72.72°C
 C) 52.5°C & 50°C
 D) 85°C & 55.72°C

Let $A \rightarrow x$
 $B \rightarrow y$

$\frac{l}{KA} = R$

$\frac{120}{11R} = \frac{y-40}{3R}$
 $y = \frac{360}{11} + 40$
 $= 32.72 + 40$
 $= 72.72^\circ\text{C}$

Same current

$\frac{100-x}{R} = \frac{100-40}{11R} \Rightarrow \frac{100-x}{R} = \frac{100-40}{11R}$
 89.1°C

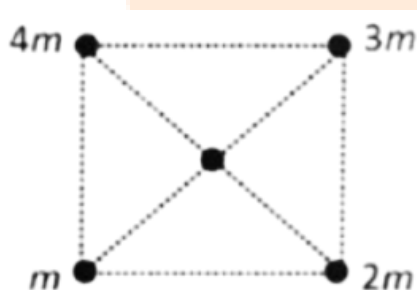
$x = 100 - \frac{120}{11}$

$4 + \frac{3}{2} = \frac{11}{2}$

$\frac{6 \times 2}{6+2} = \frac{12}{8} = \frac{3}{2}R$

Question: In the given situation force at centre on 1 kg mass is F_1 . Now if 4 m and 3 m is interchanged the force is F_2 .

Given: $\frac{F_1}{F_2} = \frac{2}{\sqrt{\alpha}}$. Find α .



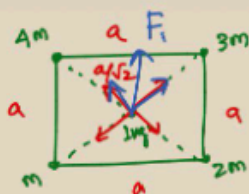
Options:

- (a) $\alpha = 5$
- (b) $\alpha = 3$
- (c) $\alpha = 2$
- (d) $\alpha = 1$

Answer: (a)

Solution:

Q-12:-



Net force due to 4m and 2m are

$$\frac{G(4m-2m) \times 1}{\left(\frac{a}{\sqrt{2}}\right)^2} \text{ (Towards 4m)}$$

Net force due to 3m and m will be

$$\frac{G(3m-m) \times 1}{\left(\frac{a}{\sqrt{2}}\right)^2} = \frac{2Gm}{\left(\frac{a^2}{2}\right)}$$

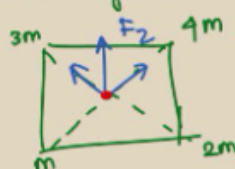
$$F_1 = \sqrt{\left(\frac{2Gm}{\frac{a^2}{2}}\right)^2 + \left(\frac{2Gm}{\frac{a^2}{2}}\right)^2}$$

$$\Rightarrow \frac{2Gm}{\frac{a^2}{2}} \sqrt{2}$$

Now, if 4m and 3m are interchanged;

net force due to 3m and 2m

will be :- $\frac{Gm}{\left(\frac{a}{\sqrt{2}}\right)^2} \Rightarrow \frac{Gm}{\frac{a^2}{2}}$



Net force due to 4m and m will be:-

$$\frac{3Gm}{\frac{a^2}{2}} \Rightarrow F_2 = \frac{6Gm}{\frac{a^2}{2}} \sqrt{10}$$

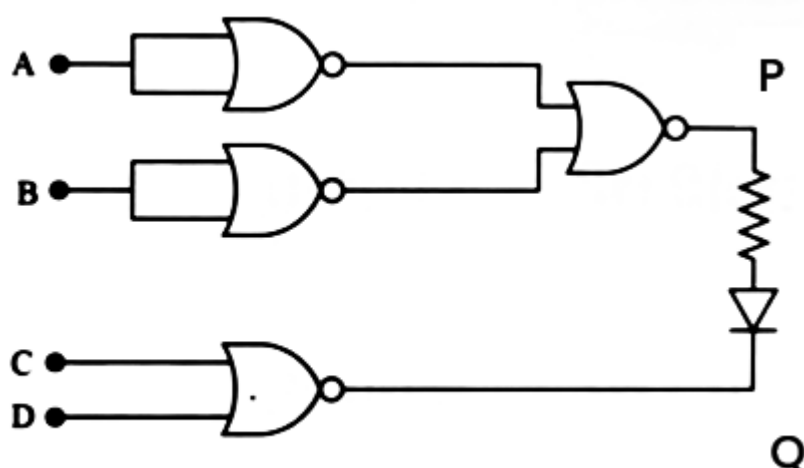
$$\frac{F_1}{F_2} = \frac{2}{\sqrt{\alpha}}$$

$$\Rightarrow \frac{\frac{2\sqrt{2}Gm}{\left(\frac{a^2}{2}\right)}}{\frac{6Gm\sqrt{10}}{\frac{a^2}{2}}} = \frac{2}{\sqrt{\alpha}}$$

$$\Rightarrow \frac{2\sqrt{2}}{\sqrt{10}} = \frac{2}{\sqrt{\alpha}}$$

$$\Rightarrow \sqrt{\frac{1}{5}} = \frac{1}{\sqrt{\alpha}} \Rightarrow \boxed{\alpha = 5} \rightarrow \text{Correct option (1)}$$

Question: In the given logic circuit (all NOR gates), for which input combination (ABCD) does the current flow?

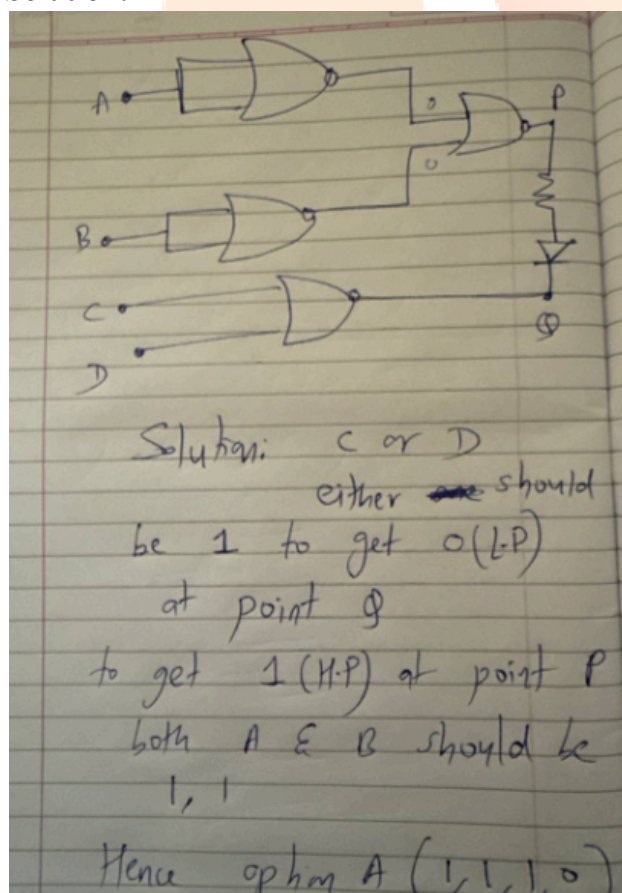


Options:

- (a) 1,1,1,0
- (b) 0,0,1,0
- (c) 1,0,0,0
- (d) 0,0,0,1

Answer: (a)

Solution:



Question: In adiabatic compression temperature of gas becomes 4 times while volume decreased $\frac{1}{2}$ times. Find γ ?

Options:

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Answer: (c)

Solution:

We have $TV^{\gamma-1} = \text{const.}$
 $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$
 $\boxed{T_2 = 4T_1} \text{ or } \frac{T_2}{T_1} = 4$
 $\boxed{V_2 = \frac{V_1}{2}} \text{ or } \frac{V_1}{V_2} = 2$
 $\rightarrow \text{So, } 4 = 2^{\gamma-1}$
 $\text{So, } 2^2 = 2^{\gamma-1}$
 $2 = \gamma - 1$
 $3 = \gamma$

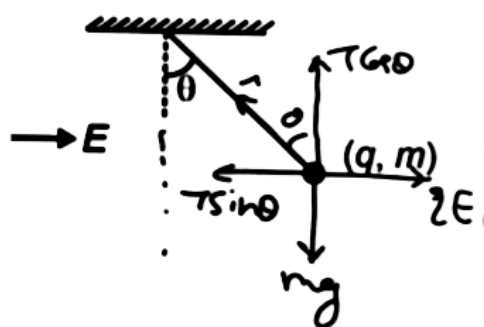
Question: A simple pendulum with bob (mass m & charge q) is in equilibrium in presence of horizontal electric field E then tension in thread is

Options:

- (a) $mg + qE$
- (b) $\sqrt{m^2g^2 + q^2E^2}$
- (c) $\sqrt{mg + qE}$
- (d) $mg + qE \tan\theta$

Answer: (b)

Solution:

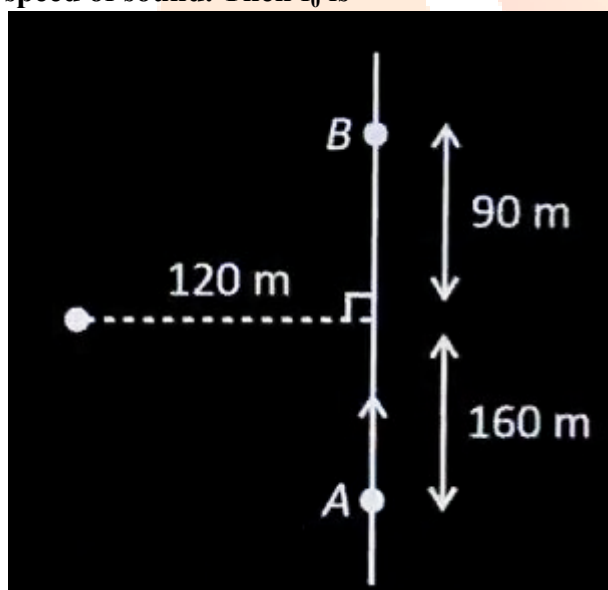


$$T \sin \theta = 2E$$

$$T \cos \theta = mg$$

$$T^2 = (2E)^2 + (mg)^2$$

Question: Detector D moves from A to B and observe the frequencies are differing by 10 Hz. Source is emitting frequency f_0 as shown: Speed of detector is 35 times less than speed of sound. Then f_0 is



Options:

- (a) 400 Hz
- (b) 250 Hz
- (c) 350 Hz
- (d) 150 Hz

Answer: (b)

Solution:

Ans:

Given; $v_0 = \frac{v}{35}$

$\tan \alpha = \frac{120}{160}$
 $\alpha = 37^\circ$
 $\tan \beta = \frac{120}{90}$
 $\beta = 53^\circ$

$$f_A = \frac{(v_0 + v_0 \cos \alpha)}{v} f_0$$

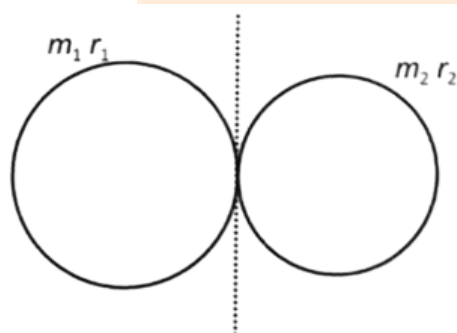
$$f_B = \left(\frac{v - v_0 \cos \beta}{v} \right) f_0$$

$f_A - f_B = 10$
 $\left(\frac{v_0 \frac{4}{5} + v_0 \frac{3}{5}}{v} \right) f_0 = 10$

$\frac{7}{5} \times \frac{1}{35} f_0 = 10$
 $f_0 = 250 \text{ Hz}$

Correct option $\rightarrow$ (3).

Question: Disk $m_1 = 5 \text{ kg}$ and radius $r_1 = 10 \text{ cm}$ and disk $m_2 = 10 \text{ kg}$ and radius $r_2 = 50 \text{ cm}$ are arranged as shown in figure. Find moment of inertia about an axis the common tangent and parallel to the plane of the disk.



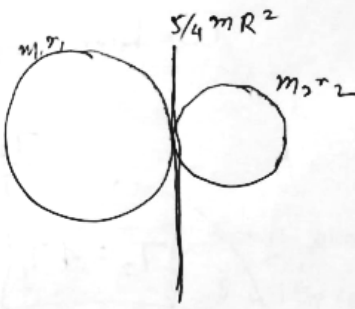
Options:

- (a) $\frac{31}{8} \text{ kg m}^2$
- (b) $\frac{57}{64} \text{ kg m}^2$
- (c) $\frac{41}{8} \text{ kg m}^2$
- (d) $\frac{51}{16} \text{ kg m}^2$

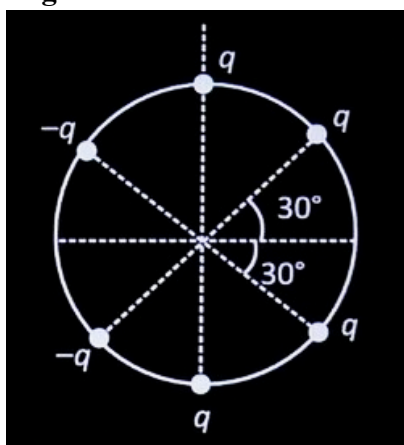
Answer: (d)

Solution:

$$I = \frac{5}{4} \left(5 \times \frac{1}{100} + \frac{10 \times 1}{4} \right)$$

$$= \frac{1}{20} + \frac{10}{4} = \frac{51}{16}$$


Question: Six charges (four +q, two -q) are present at circle of radius r and centred at origin as shown. Electric field at origin is



Options:

- (a) $\frac{\sqrt{3}q}{4\pi \epsilon_0 r^2} \hat{i}$
- (b) $\frac{\sqrt{3}q}{4\pi \epsilon_0 r^2} (-\hat{i})$
- (c) $\frac{\sqrt{3}q}{2\pi \epsilon_0 r^2} (-\hat{i})$
- (d) $\frac{\sqrt{3}q}{\pi \epsilon_0 r^2} (\hat{i})$

Answer: (c)

Solution:

E_{net} due to charge 1, 2, 4, 5 will be:-

$$\frac{\sqrt{3}kq}{r^2} \Rightarrow \frac{\sqrt{3} \times q}{2\pi\epsilon_0 r^2} (i)$$

E_{net} due to 3, 6 charge will be 0

$$E_{\text{origin}} = \frac{\sqrt{3}q}{2\pi\epsilon_0 r^2} (-i)$$

Question: An α particle with K.E. 7.9 MeV is projected towards a stationary tangent nucleus of $Z = 79$. Find the distance of closest approach.

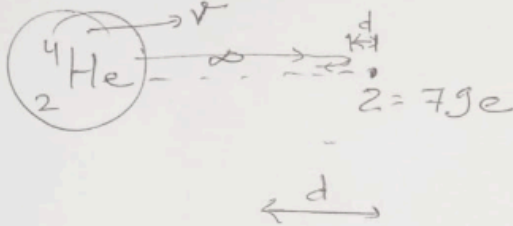
Options:

- (a) 1.44×10^{-14}
- (b) 2.88×10^{-14}
- (c) 1.44×10^{-15}
- (d) 2.88×10^{-15}

Answer: (b)

Solution:

Soln



$$7.9 \times 10^6 \times (1.6 \times 10^{-19}) = \frac{k(2e)(79e)}{d}$$

$$\Rightarrow d = \frac{k(2e) \times 79}{7.9 \times 10^6} \Rightarrow d = 2.88 \times 10^{-14}$$

Question: The energy requires to excite electron from first Bohr's orbit of Hydrogen atom to second Bohr's orbit in J is:

Options:

- (a) 1.634×10^{18} J
- (b) 1.2×10^{-18} J
- (c) 0.2×10^{-18} J
- (d) 1.2×10^{-20} J

Answer: (c)

Solution:

$$\left. \begin{array}{l} E_1 = -\frac{13.6 \times 1}{1} \\ E_2 = -\frac{13.6}{4} \end{array} \right\} \Delta E = -13.6 \times \frac{3}{4} \times 10^{-19} \times 1.6 \times 10^{-19} = 10.2 \text{ eV}$$

$$\begin{aligned} & -13.6 \times \frac{3}{4} \times 1.6 \times 10^{-19} \\ & 16.32 \times 10^{-19} \\ & = 1.632 \times 10^{-18} \end{aligned}$$

Question: A photon is incident on particle having mass $m = 15.356$ amu. What should be the frequency of photon so that particle of mass 'm' breaks into four α -particles:

(given: $m_\alpha = 4.004$ amu; $h = 6.6 \times 10^{-34}$ Js)

Options:

- (a) 14.9×10^{19} kHz
- (b) 12.9×10^{19} kHz
- (c) 9.9×10^{19} kHz
- (d) 10.9×10^{19} kHz

Answer: (a)

Solution:

Total mass of products
 $4 \times 4.004 = 16.016 \text{ amu}$

Mass defect
 $\Delta m = 16.016 - 15.356$
 $= 0.660 \text{ amu}$

converted to kg;
 $\Delta m = 0.660 \times 1.66 \times 10^{-27}$
 $= 1.0956 \times 10^{-27} \text{ kg}$

$E = \Delta mc^2$
 $= 1.0956 \times 10^{-27} \times c^2$
 $= 9.86 \times 10^{-11} \text{ J}$

$E = h\nu$
 $\nu = \frac{E}{h} = \frac{9.86 \times 10^{-11}}{6.6 \times 10^{-34}}$
 $= 1.49 \times 10^{23} \text{ Hz}$
 $= 14.9 \times 10^{19} \text{ KHz}$

Question: A closed tube filled with ideal gas is rotating with ' ω ' along axis passing through end A. Find pressure at other end B:

Given pressure at 'A' is P_A :

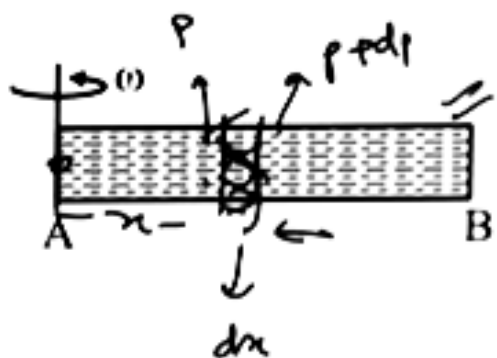


Options:

- (a) $P_A e^{\frac{\omega^2 l^2 m}{2RT}}$
- (b) $P_A e^{\frac{\omega^2 l^2 m}{RT}}$
- (c) $P_A e^{\frac{\omega^2 l^2 m}{3RT}}$
- (d) $P_A e^{\frac{\omega^2 l^2 m}{4RT}}$

Answer: (a)

Solution:



$$(dP)A = (dm)\omega^2 x$$

$$(dP)A = P(A dx) \omega^2 x$$

$$\frac{dP}{P} = \frac{M}{R_T} \omega^2 \int_0^l x dx$$

$$\ln\left(\frac{P_B}{P_A}\right) = \frac{m\omega^2 l^2}{2RT}$$

$$P_B = P_A e^{\frac{m\omega^2 l^2}{2RT}}$$

Question: If potential is $v = 500$ volts at $(10, 20)$ and electric field given

$\vec{E} = 10x\hat{i} + 5y\hat{j} \text{ N/C}$. Find potential at origin.

Options:

- (a) 1000 volt
- (b) 2000 volt
- (c) 1500 volt
- (d) 3000 volt

Answer: (b)

Solution:

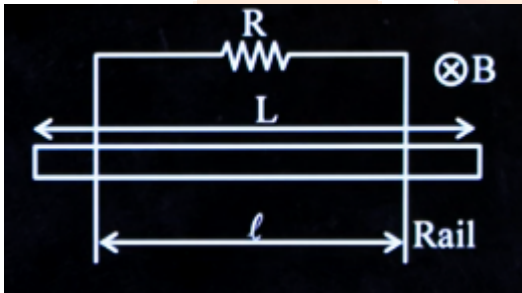
$$\int dV = - \int \underline{E} \cdot d\underline{r} \rightarrow \frac{dx}{1} + \frac{dy}{2}$$

$$\int_{V_0}^{500} dV \Rightarrow - \int_0^{10} 10x dx + \int_0^{20} 5y dy$$

$$500 - V_0 \Rightarrow - \left(500 + 1000 \right)$$

$$V_0 = 2000$$

Question: A rod of mass 'm' and length 'L' is released on a rail placed in uniform magnetic field B. What will be the terminal velocity of rod :



Options:

- (a) $\frac{mgR}{B^2 L^2}$
- (b) $\frac{mgR}{B^2 l^2}$
- (c) $\frac{Bl^2}{mg}$
- (d) $\frac{B^2 l^2 R}{mg}$

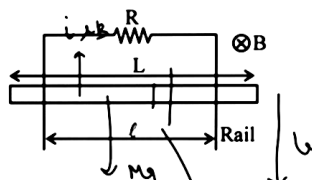
Answer: (b)

Solution:

$$mg = i l B$$

$$mg = \frac{B^2 l^2 v}{R}$$

$$v = \frac{mgR}{B^2 l^2}$$



$$i = \frac{v B l}{R}$$

JEE-Main-22-01-2026 (Memory Based)
[MORNING SHIFT]
Chemistry

Question: Arrange the given metal ions in number in increasing order of unpaired electrons in low spin complex formed by Mn^{3+} , Cr^{3+} , Fe^{3+} , Co^{3+}

Options:

- (a) $\text{Co}^{3+} < \text{Fe}^{3+} < \text{Mn}^{3+} < \text{Cr}^{3+}$
(b) $\text{Co}^{3+} < \text{Mn}^{3+} < \text{Fe}^{3+} < \text{Cr}^{3+}$
(c) $\text{Cr}^{3+} < \text{Mn}^{3+} < \text{Cr}^{3+} < \text{Fe}^{3+}$
(d) $\text{Cr}^{3+} < \text{Mn}^{3+} < \text{Co}^{3+} < \text{Fe}^{3+}$

Answer: (a)

Question: Match the following and choose the correct option.

List- I	List-II
a) $[\text{Ag}(\text{NH}_3)_2]^+$	i) Fehling's solution
b) Zn-Hg/HCl	ii) Clemmenson's reduction
c) $\text{NH}_2 - \text{NH}_2/\text{KOH}$	iii) Tollen's reagent
d) $\text{Cu}^{2+}/\text{OH}^-$	iv) Wolff-Kishner reduction

Options:

- (a) a(i), b(ii), c(iii), d(iv)
(b) a(iv), b(iii), c(ii), d(i)
(c) a(iii), b(ii), c(iv), d(i)
(d) a(i), b(ii), c(iv), d(iii)

Answer: (c)

Question: Find the energy to excite electron from first Bohr orbit of hydrogen atom to 2nd Bohr's orbit. (in eV)

Options:

- (a) 10.2 eV
(b) 20.2 eV
(c) 15.1 eV
(d) 13.5 eV

Answer: (a)

Question: Statement-I: Sucrose is dextrorotatory and upon hydrolysis it becomes laevorotatory.

Statement-II: Sucrose on hydrolysis gives glucose and fructose such that laevorotation of glucose is more than dextrorotation of fructose.

Options:

- (a) Both Statement-I and Statement-II are correct
- (b) Both Statement-I and Statement-II are incorrect
- (c) Statement-I is correct, Statement-II is incorrect
- (d) Statement-II is correct, Statement-I is incorrect

Answer: (c)

Question: Which of the following is the correct order of the reactivity of given nucleophiles when treated with CH_3Br in methanol?

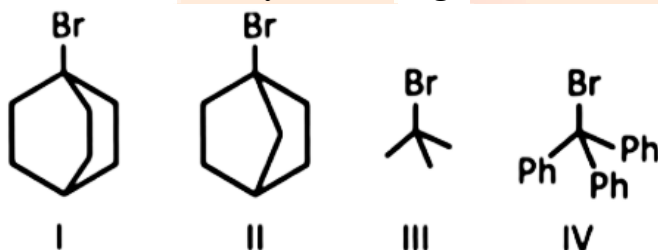
F^- , I^- , $\text{C}_2\text{H}_5\text{O}^-$, $\text{C}_6\text{H}_5\text{O}^-$

Options:

- (a) $\text{I}^- > \text{C}_2\text{H}_5\text{O}^- > \text{C}_6\text{H}_5\text{O}^- > \text{F}^-$
- (b) $\text{I}^- > \text{F}^- > \text{C}_2\text{H}_5\text{O}^- > \text{C}_6\text{H}_5\text{O}^-$
- (c) $\text{I}^- > \text{C}_2\text{H}_5\text{O}^- > \text{F}^- > \text{C}_6\text{H}_5\text{O}^-$
- (d) $\text{C}_6\text{H}_5\text{O}^- > \text{F}^- > \text{I}^- > \text{C}_2\text{H}_5\text{O}^-$

Answer: (a)

Question: Reactivity of following on the basis of $\text{S}_{\text{N}}1$ mechanism.



Options:

- (a) $\text{IV} > \text{III} > \text{I} > \text{II}$
- (b) $\text{II} > \text{IV} > \text{I} > \text{I}$
- (c) $\text{III} > \text{IV} > \text{I} > \text{II}$
- (d) $\text{IV} > \text{III} > \text{II} > \text{I}$

Answer: (a)

Question: Given below are two statements.

Statement-I: HX bond length is higher in HCl than HF.

Statement-II: The lowest boiling point in hydride of group 15 element is having covalency 4.

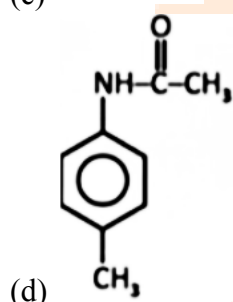
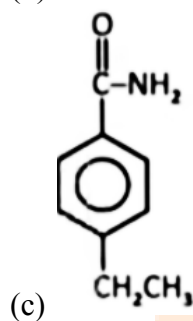
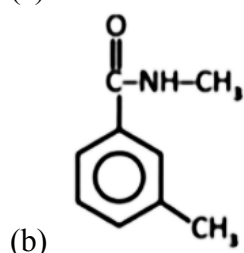
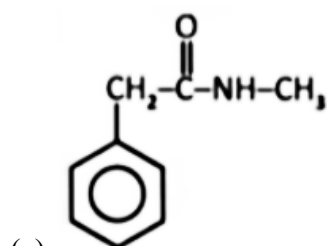
Options:

- (a) Both Statement-I and Statement-II is correct
- (b) Both Statement-I and Statement-II is incorrect
- (c) Statement-I is correct but statement-II is incorrect
- (d) Statement-I is incorrect but statement-II is correct

Answer: (c)

Question: A compound 'A' with molecular formula $\text{C}_9\text{H}_{11}\text{NO}$ reacts with Br_2/NaOH to give (X). (X) on reaction with NaNO_2 in dil. HCl gives compounds (Y). When (Y) is treated with CuCN , followed by hydrolysis gives (Z). The compound (A) on hydrolysis also gives compound (Z). Identify compound (A)

Options:



Answer: (c)

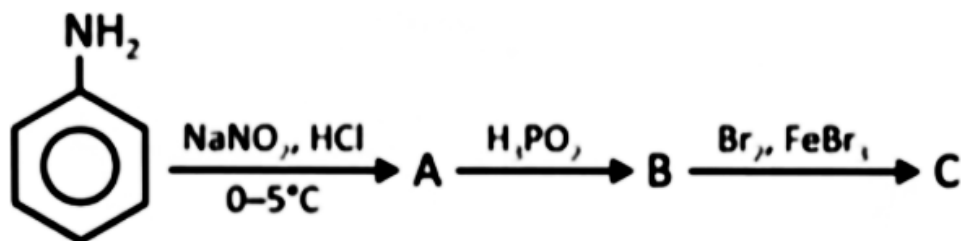
Question: Which of the following statements is correct regarding the nature and directive influence of $-\text{NO}_2$ group in nitration of benzene.

Options:

- (a) It is an activating group and ortho/para director
- (b) It is a deactivating group and ortho/para director
- (c) It is a deactivating group and meta director
- (d) It is an activating group and meta director

Answer: (c)

Question: Consider the following sequence of reaction and identify A, B and C respectively.



Options:

- (a) $\text{C}_6\text{H}_5\text{OH}$, C_6H_6 , $\text{C}_6\text{H}_4\text{Br}_2$
- (b) $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$, C_6H_6 , $\text{C}_6\text{H}_5\text{Br}$
- (c) $\text{C}_6\text{H}_5\text{NO}_2$, $\text{C}_6\text{H}_5\text{OH}$, $\text{C}_6\text{H}_4\text{Br}$
- (d) $\text{C}_6\text{H}_5\text{Cl}$, $\text{C}_6\text{H}_5\text{OH}$, C_6H_6

Answer: (b)

Question: Statement-I: Ortho & Para nitro phenol can be differentiated by steam distillation.

Statement-II: Glycerol is separated from spent lie by distillation under reduced pressure

Statement-III: Chromatography separation based on differential affinities of components for a Stationary phase.

Statement IV: Aniline is commonly separated from a mixture of water by crystallization.

Select correct options

Options:

- (a) Only I & IV
- (b) Only I, II & III
- (c) Only I, III
- (d) All of these

Answer: (b)

Question: Given below are two statements

Statement-I: K_H is constant with change in concentration of gas till solution is dilute at given temperature.

Statement-II: According to Henry's Law, partial pressure of gas in vapour phase is inversely proportional to mole fraction of gas in solution.

Options:

- (a) Both Statement-I and Statement-II are correct
- (b) Both Statement-I and Statement-II are incorrect
- (c) Statement-I is correct, Statement-II is incorrect
- (d) Statement-I is incorrect and Statement-II is correct

Answer: (c)

Question: Consider a first order reaction:

$\text{A} \rightarrow \text{Products}$

3 different solutions are taken rate of reaction

Solution 1: 100 mL 10M 'A' - r_1

Solution 2: 200 mL 10M 'A' - r_2

Solution 3: 100 mL 10M 'A' + 100mL water - r_3

The correct order of the rate of reactions is,

Options:

- (a) $r_1 = r_2 = r_3$

- (b) $r_1 = r_2 < r_3$
 (c) $r_1 = r_2 > r_3$
 (d) $r_1 < r_2 < r_3$

Answer: (c)

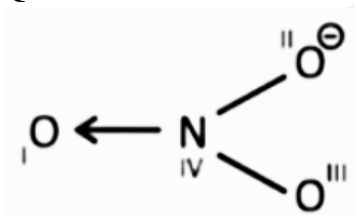
Question: Bohr's radius of H-atom is 2.12×10^{-10} m. Calculate the energy at this level.

Options:

- (a) -5.44×10^{-19} J
 (b) -2.176×10^{-18} J
 (c) -54.4×10^{-19} J
 (d) -2.3×10^{-19} J

Answer: (a)

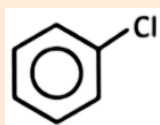
Question: Find the formal charge of N^{IV} and O^I, O^{II}, O^{III}

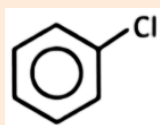


Options:

- (a) 0, +1, -1, +2
 (b) +1, -1, -1, 0
 (c) -1, 0, +2, +1
 (d) +1, -1, 0, -1

Answer: (b)



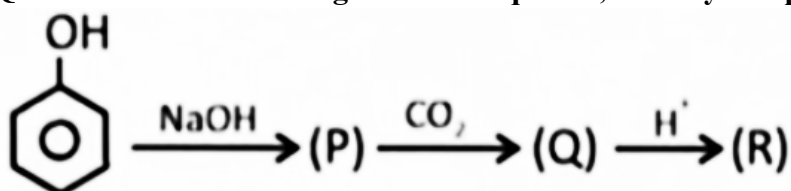
Question: For , the incorrect statement is, 'P'

Options:

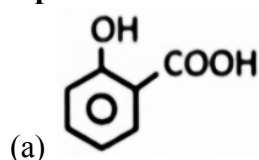
- (a) 'P' is less reactive than benzyl chloride towards nucleophilic substitution reaction.
 (b) In 'P' C-Cl bond has partial double bond character
 (c) 'Cl' is an ortho-para directing group towards electrophilic aromatic substitution
 (d) 'P' can undergo nucleophilic substitution reaction at normal conditions

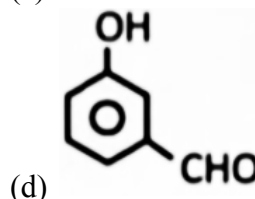
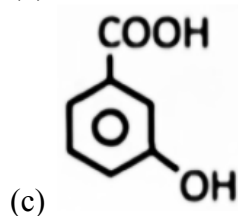
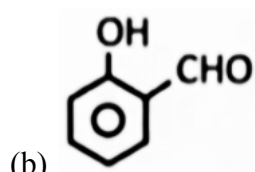
Answer: (d)

Question: In the following reaction sequence, identify compound (R)



Options:





Answer: (a)

Question: Which of the following statement is correct regarding element having atomic number 79.

Options:

- (a) It's first ionisation enthalpy is maximum in its group
- (b) It's first ionisation enthalpy is minimum in its group
- (c) It belongs to group 10 of periodic table
- (d) It belongs to 5th period of periodic table

Answer: (a)

Question: An element from 1st transition series and another element of 3rd transition series (same group) do not liberate H₂ gas from dilute acids like HCl. Both form halides. The hybridisation state of metal ion halide respectively are

Options:

- (a) Both sp³
- (b) Both dsp²
- (c) sp³ and dsp²
- (d) dsp² and sp³

Answer: (c)

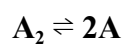
Question: Sodium extract of organic compound of 0.1 g is treated with chlorine water and CCl₄ which dissolves in organic solvent produce a violet colour upon treatment with AgNO₃ a yellow ppt of 0.12 g is produce. Calculate the percentage of Halide in organic compound.

Options:

- (a) 65
- (b) 91
- (c) 34
- (d) 80

Answer: (a)

Question: For the reaction given below at 25°C



Find $\ln K_p$

Given $(\Delta G^\circ)_A = -50.384 \text{ KJ/mol}$

Given $(\Delta G^\circ_f)_{A_2} = -100 \text{ KJ/mol}$

Options:

(a) 0.43

(b) 0.23

(c) 0.31

(d) 0.53

Answer: (c)



JEE-Main-22-01-2026 (Memory Based)
[MORNING SHIFT]
Maths

Question: If sum of first 4 terms of an A.P is 6 and sum of first 6 terms is 4, then sum of first 12 terms of an A.P is

Options:

- (a) -22
- (b) -21
- (c) -23
- (d) -24

Answer: (a)

$$S_4 = 6$$

$$S_6 = 4$$

$$\text{Find } S_{12} = ?$$

$$S_4 = 6 \Rightarrow \frac{4}{2} [2a + 3d] = 6$$

$$\Rightarrow 2a + 3d = 3 \dots\dots\dots(1)$$

$$S_6 = 4 \Rightarrow \frac{6}{2} [2a + 5d] = 4$$

$$\Rightarrow 2a + 5d = \frac{4}{3} \dots\dots\dots(2)$$

$$Eq(2) - Eq(1)$$

$$2d = -\frac{5}{3}$$

$$\Rightarrow d = \frac{-5}{6}$$

$$\text{by (1)} \Rightarrow 2a = 3 - 3d$$

$$2a = 3 - 3\left(\frac{-5}{6}\right)$$

$$2a = \frac{11}{2}$$

$$S_{12} = \frac{12}{2} [2a + 11d]$$

$$= 6 \left[\frac{11}{2} + 11\left(\frac{-5}{6}\right) \right]$$

$$= 6 \left[\frac{33-55}{6} \right]$$

$$= -22$$

Question: The coefficient of x^{48} in $1.(1+x) + 2.(1+x)^2 + 3.(1+x)^3 + \dots\dots\dots + 100.(1+x)^{100}$ is

Options:

- (a) $(^{101}C_{46}) - 100$
- (b) $100(^{101}C_{49}) - ^{101}C_{50}$
- (c) $100(^{101}C_{46}) - ^{101}C_{47}$
- (d) $^{101}C_{47} - ^{101}C_{46}$

Answer: (b)

$$\begin{aligned}
 s &= y + 2y^2 + 3y^3 + \dots + 100y^{100}, \quad y = 1 + x \\
 ys &= y^2 + 2y^3 + \dots + 100y^{101} \\
 (1 - y)s &= (y + y^2 + y^3 + \dots + y^{100}) - 100y^{101} \\
 (1 - y)s &= \frac{y(1 - y^{100})}{1 - y} - 100y^{101} \\
 S &= \frac{y - y^{101}}{(1 - y)^2} - \frac{100y^{101}}{1 - y} = \frac{(1 + x) - (1 + x)^{101}}{x^2} + \frac{100(1 + x)}{x} \\
 \text{coeff of } x^{48} &= -^{101}C_{50} + 100 \cdot ^{101}C_{49}
 \end{aligned}$$

Question: If $A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$, then the value of $|A^{2025} - 3A^{2024} + A^{2023}|$ is

Answer: (16)

$$\begin{aligned}
 A &= \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} \\
 A^2 &= \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} = \begin{bmatrix} 13 & 21 \\ 21 & 34 \end{bmatrix} \\
 |A| &= 1 \\
 |A^{2025} - 3A^{2024} + A^{2023}| &= |A^{2025}(A^2 - 3A + I)| \\
 &= |A^{2025}| |A^2 - 3A + I| \\
 &= |A|^{2025} |A^2 - 3A + I| \\
 &= (1)^{2025} \begin{vmatrix} 8 & 12 \\ 12 & 20 \end{vmatrix} \\
 &= 1(160 - 144) \\
 &= 16
 \end{aligned}$$

Question: If the domain of the function $\frac{1}{\ln(10 - x)} + \sin^{-1}\left(\frac{x + 2}{2x + 3}\right)$ is $(-\infty, -a] \cup (-1, b) \cup (b, c)$, then $(b + c + 3a)$ is equal

Options:

- (a) 22
- (b) 24
- (c) 23
- (d) 21

Answer: (b)

$$\begin{aligned}
 10 - x &> 0, \quad 10 - x \neq 1, \quad -1 \leq \frac{x + 2}{2x + 3} \leq 1 \\
 x < 10, \quad x \neq 9, \quad \frac{x + 2}{2x + 3} + 1 &\geq 0, \quad \frac{x + 2}{2x + 3} - 1 \leq 0 \\
 \Rightarrow \frac{3x + 5}{2x + 3} &\geq 0, \quad \frac{-x - 1}{2x + 3} \leq 0 \\
 \Rightarrow x &\in \left(-\infty, -\frac{5}{3}\right) \cup \left(-\frac{3}{2}, \infty\right), \quad x \in \left(-\infty, -\frac{3}{2}\right) \cup [-1, \infty) \\
 \text{Domine} &= \left(-\infty, -\frac{5}{3}\right] \cup [-1, 9) \cup (9, 10) \\
 a &= \frac{5}{3}, \quad b = 9, \quad c = 10 \\
 b + c + 3a &= 19 + 5 = 24
 \end{aligned}$$

Question: If $\tan^{-1}(4x) + \tan^{-1}(6x) = \frac{\pi}{6}$ then no of solution in $\left(-\frac{1}{2\sqrt{6}}, \frac{1}{2\sqrt{6}}\right)$

Answer: (1)

$$\frac{4x+6x}{1-14x^2} = \frac{1}{\sqrt{3}}$$

$$10\sqrt{3}x = 1 - 24x^2$$

$$24x^2 + 10\sqrt{3}x - 1 = 0 \Rightarrow x = \frac{-10\sqrt{3} \pm \sqrt{300+96}}{48}$$

$$\Rightarrow x = \frac{-10\sqrt{3} + \sqrt{396}}{48} = \frac{2}{\sqrt{396} + \sqrt{360}}$$

No of solutions = 1

Question: Let $M = \{1, 2, 3, \dots, 16\}$ and R be a relation on M defined by xRy if and only if $4y = 5x - 3$. Then, the number of elements required to added in R to make it symmetric is

Options:

- (a) 2
- (b) 3
- (c) 4
- (d) 5

Answer: (a)

$$M = \{1, 2, 3, \dots, 16\}$$

$$xRy \Leftrightarrow 4y = 5x - 3$$

$$\Rightarrow y = \frac{5x-3}{4}$$

$$\Rightarrow R = \{(3, 3), (7, 8), (11, 13)\}$$

To make R symmetric, we need to add $(8, 7)$

& $(13, 11)$

$\therefore$ 2 elements are required.

Question: The solution of the differential equation $xdy - ydx = \sqrt{x^2 + y^2} dx$ is (where c is integration constant)

Options:

(a) $\sqrt{x^2 + y^2} = cx^2 - y$

(b) $\sqrt{x^2 + y^2} = cx^2 + y$

(c) $\sqrt{x^2 + y^2} = cx - y$

(d) $\sqrt{x^2 + y^2} = cx + y$

Answer: (a)

$$\frac{xdy-ydx}{x^2} = \sqrt{x^2 + y^2}$$

$$\int \frac{d\frac{y}{x}}{\sqrt{1 + \left(\frac{y}{x}\right)^2}} = \int \frac{dx}{x}$$

$$\log\left(\frac{y}{x} + \sqrt{1 + \left(\frac{y}{x}\right)^2}\right) = \ln x + \log_c$$

$$xc = \frac{y}{x} + \frac{\sqrt{y^2+x^2}}{x}$$

$$x^2c = y + \sqrt{y^2 + x^2}$$

Question: The number of real solution of equation $x | x + 4 | + 3 | x + 2 | + 10 = 0$ is/are :
Options:

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Answer: (a)

Case I: $x < -4$

$$-x(x+4) - 3(x+2) + 10 = 0$$

$$-x^2 - 7x + 4 = 0 \Rightarrow x^2 + 7x - 4 = 0$$

$$x = \frac{-7 \pm \sqrt{49+16}}{2} = \frac{-7 \pm \sqrt{65}}{2}$$

$$x = \frac{-7 - \sqrt{65}}{2}$$

Case II : $-4 \leq x < -2$

$$x(x+4) - 3(x+2) + 10 = 0$$

$$x^2 + x + 4 = 0$$

Case III: $x \geq -2$

$$x(x+4) + 3x + 6 + 10 = 0$$

$$x^2 + 7x + 16 = 0$$

No. of solutions = 1

Question: The value of $\frac{\cos^2 48^\circ - \cos^2 12^\circ}{\sin^2 24^\circ - \sin^2 6^\circ}$ is $\frac{\alpha + \beta\sqrt{5}}{\gamma}$ Find $\alpha + \beta + \gamma$
Answer: ($\alpha + \beta + \gamma = 12$)

$$\begin{aligned} & \frac{\cos^2 48 - \sin^2 12}{\sin^2 24 - \sin^2 6} \\ &= \frac{\cos 60 \cos 36}{\sin 30 \sin 18} \\ &= \frac{\cos 36}{\sin 18} \\ &= \frac{\frac{1+\sqrt{5}}{4}}{\frac{-1+\sqrt{5}}{4}} \\ &= \frac{\sqrt{5}+1}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1} \\ &= \frac{5+1+2\sqrt{5}}{5-1} = \frac{6+2\sqrt{5}}{4} \end{aligned}$$

$$\cos^2 a - \sin^2 b = \cos(a+b) \cos(a-b)$$

$$\alpha + \beta + \gamma = 12$$

Question: The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{dx}{[x]+4}$ is Where $[.]$ denotes greatest integer function.
Options:

- (a) $\frac{\pi}{20} + \frac{7}{20}$
- (b) $\frac{20}{7\pi} - \frac{60}{7}$
- (c) $\frac{20}{7\pi} - \frac{60}{1}$
- (d) $\frac{20}{7\pi} + \frac{1}{60}$

Answer: (d)

$$\begin{aligned} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{dx}{[x]+4} &= \int_{-\frac{\pi}{2}}^{-1} \frac{dx}{2} + \int_{-1}^0 \frac{dx}{3} + \int_0^1 \frac{dx}{4} + \int_1^{\frac{\pi}{2}} \frac{dx}{5} \\ &= \frac{1}{2} \left(-1 + \frac{\pi}{2} \right) + \frac{1}{3} (0 + 1) + \frac{1}{4} + \frac{1}{5} \left(\frac{\pi}{2} - 1 \right) \\ &= -\frac{1}{2} + \frac{\pi}{4} + \frac{1}{3} + \frac{1}{4} + \frac{\pi}{10} - \frac{1}{5} \\ &= \frac{7}{12} - \frac{1}{2} - \frac{1}{5} + \frac{\pi}{4} + \frac{\pi}{10} \\ &= \frac{35-30-6}{60} + \frac{5\pi+2\pi}{20} = -\frac{1}{60} + \frac{7\pi}{20} \end{aligned}$$

Question: If a line $ax + y = 1$ does not intersect the hyperbola $x^2 - 9y^2 = 9$ then α possible value of α is :

Options:

- (a) 0.2

(b) 0.3

(c) 0.4

(d) 0.5

Answer: (d)

$$x^2 - 9(1 - \alpha x)^2 = 9$$

$$x^2 - 9(1 + \alpha^2 x^2 - 2\alpha x) = 9$$

$$(1 - 9\alpha^2)x^2 + 18\alpha x - 18 = 0$$

$$D < 0$$

$$(18\alpha)^2 + 4 \times 18(1 - 9\alpha^2) < 0$$

$$\Rightarrow 18\alpha^2 + 4 - 36\alpha^2 < 0 \Rightarrow 4 < 18\alpha^2$$

$$\Rightarrow \alpha^2 > \frac{2}{9} = 0.222$$

Correct ans (4)

Question: If

$$\int (\cos x)^{-5/2} (\sin x)^{-11/2} dx = \frac{p_1}{q_1} (\cot x)^{9/2} + \frac{p_2}{q_2} (\cot x)^{1/2} - \frac{p_4}{q_4} (\cot x)^{-3/2} + c$$

$$\frac{15p_1p_2p_3p_4}{q_1q_2q_3q_4}$$

(where c is constant of integration), then value of $\frac{15p_1p_2p_3p_4}{q_1q_2q_3q_4}$ is

Options:

(a) 14

(b) 16

(c) 10

(d) 9

Answer: (b)

$$\int (\cos x)^{-\frac{5}{2}} (\sin x)^{-\frac{11}{2}} dx$$

$$I = \int \frac{1}{\cos^{\frac{5}{2}} x \sin^{\frac{11}{2}} x} dx$$

$$I = \int \frac{\operatorname{cosec}^8 x}{\cos^{\frac{5}{2}} x} dx$$

$$\text{let } \cot x = t$$

$$-\operatorname{cosec}^2 x dx = dt$$

$$\therefore I = - \int \frac{(1+t^2)^3}{t^{\frac{5}{2}}} dt$$

$$= - \int \frac{(1+t^6+3t^2+3t^4)dt}{t^{\frac{5}{2}}}$$

$$= - \int \left(t^{-\frac{5}{2}} + 3t^{\frac{3}{2}} + 3t^{-\frac{1}{2}} + t^{\frac{7}{2}} \right) dt$$

$$= - \left[\frac{t^{\frac{9}{2}}}{\frac{9}{2}} + \frac{3t^{\frac{5}{2}}}{\frac{5}{2}} + \frac{3t^{\frac{1}{2}}}{\frac{1}{2}} + \frac{t^{\frac{9}{2}}}{\frac{9}{2}} \right]$$

$$= -\frac{2}{9} \cot^{\frac{9}{2}} x - \frac{6}{5} \cot^{\frac{5}{2}} x - 6 \cot^{\frac{1}{2}} x + \frac{2}{3} \cot^{-\frac{3}{2}} x + c$$

$$\therefore \frac{p_1}{q_1} = \frac{-2}{9}, \frac{p_2}{q_2} = \frac{-6}{5}, \frac{p_3}{q_3} = -6, \frac{p_4}{q_4} = -\frac{2}{3}$$

$$\therefore \frac{15p_1p_2p_3p_4}{q_1q_2q_3q_4} = 16$$

Question: Let $6 \int_1^x f(t)dt = 3xf(x) - x^3 + 4$, then find $f(2) - f(3)$

Answer: (-3)

$$x = 1$$

$$0 = 3f(1) + 3$$

$$f(1) = -1$$

$$6f(x) = 3f(x) + 3xf'(x) - 3x^2$$

$$\Rightarrow 3xf'(x) - 3f(x) - 3x^2 = 0$$

$$\Rightarrow x \frac{dy}{dx} - y = x^2, y = f(x)$$

$$\Rightarrow \frac{x dy - y dx}{x^2} = 1$$

$$\Rightarrow \frac{y}{x} = x + c \Rightarrow f(x) = x^2 + cx$$

$$f(1) = 1 + c \Rightarrow c = -2$$

$$f(x) = x^2 - 2x$$

$$f(2) - f(3) = (4 - 4) - (9 - 6)$$

$$= -3$$